Запуск ядра с init файлом

1. Очистка от старых записей:

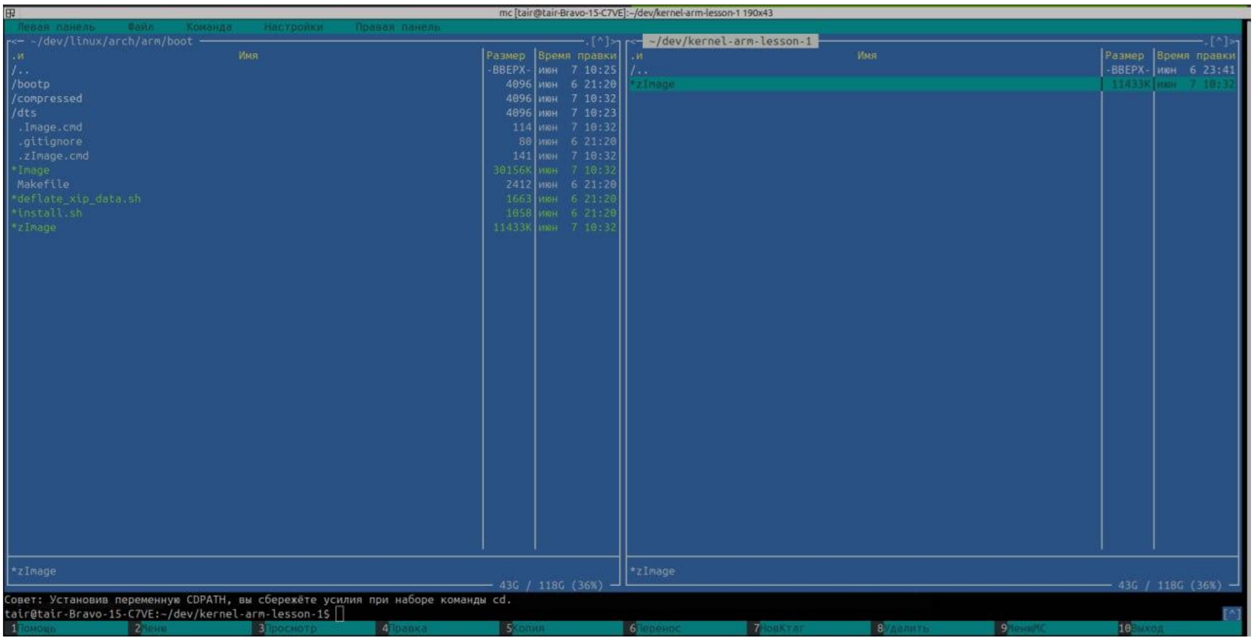
```
tair@tair-Bravo-15-C7VE:~/dev/linux$ make mrproper
  CLEAN   certs
  CLEAN   drivers/firmware/efi/libstub
  CLEAN   drivers/scsi
  CLEAN   drivers/tty/vt
  CLEAN   init
  CLEAN   lib/crc
  CLEAN   lib
  CLEAN   usr
  CLEAN   .
  CLEAN   vmlinux.symvers modules.builtin modules.builtin.modinfo .vmlinux.export.c
  CLEAN   scripts/basic
  CLEAN   scripts/dtc
  CLEAN   scripts/kconfig
  CLEAN   scripts/mod
  CLEAN   scripts
  CLEAN   include/config include/generated .config .config.old .version
tair@tair-Bravo-15-C7VE:~/dev/linux$
```

2. Создание дефолтного конфига:

```
tair@tair-Bravo-15-C7VE:~/dev/linux$ ARCH=arm make defconfig
  HOSTCC  scripts/basic/fixdep
  HOSTCC  scripts/kconfig/conf.o
  HOSTCC  scripts/kconfig/confdata.o
  HOSTCC  scripts/kconfig/expr.o
  LEX     scripts/kconfig/lexer.lex.c
  YACC    scripts/kconfig/parser.tab.[ch]
  HOSTCC  scripts/kconfig/lexer.lex.o
  HOSTCC  scripts/kconfig/menu.o
  HOSTCC  scripts/kconfig/parser.tab.o
  HOSTCC  scripts/kconfig/preprocess.o
  HOSTCC  scripts/kconfig/symbol.o
  HOSTCC  scripts/kconfig/util.o
  HOSTLD  scripts/kconfig/conf
*** Default configuration is based on 'multi_v7_defconfig'
#
# configuration written to .config
#
```

3. Сборка ядра (результат):

```
  LD      vmlinux.o
  OBJCOPY modules.builtin.modinfo
  GEN     modules.builtin
  MODPOST vmlinux.symvers
  CC      .vmlinux.export.o
  UPD     include/generated/utsversion.h
  CC      init/version-timestamp.o
  KSYMS   .tmp_vmlinux0.kallsyms.S
  AS      .tmp_vmlinux0.kallsyms.o
  LD      .tmp_vmlinux1
  NM      .tmp_vmlinux1.syms
  KSYMS   .tmp_vmlinux1.kallsyms.S
  AS      .tmp_vmlinux1.kallsyms.o
  LD      .tmp_vmlinux2
  NM      .tmp_vmlinux2.syms
  KSYMS   .tmp_vmlinux2.kallsyms.S
  AS      .tmp_vmlinux2.kallsyms.o
  LD      vmlinux
  NM      System.map
  SORTTAB vmlinux
  OBJCOPY arch/arm/boot/Image
  Kernel: arch/arm/boot/Image is ready
  LDS     arch/arm/boot/compressed/vmlinux.lds
  AS      arch/arm/boot/compressed/head.o
  GZIP    arch/arm/boot/compressed/piggy_data
  CC      arch/arm/boot/compressed/misc.o
  CC      arch/arm/boot/compressed/decompress.o
  CC      arch/arm/boot/compressed/string.o
  AS      arch/arm/boot/compressed/hyp-stub.o
  CC      arch/arm/boot/compressed/fdt_rw.o
  CC      arch/arm/boot/compressed/fdt_ro.o
  CC      arch/arm/boot/compressed/fdt_wip.o
  CC      arch/arm/boot/compressed/fdt.o
  CC      arch/arm/boot/compressed/atags_to_fdt.o
  CC      arch/arm/boot/compressed/fdt_check_mem_start.o
  AS      arch/arm/boot/compressed/lib1funcs.o
  AS      arch/arm/boot/compressed/ashldi3.o
  AS      arch/arm/boot/compressed/bswapsdi2.o
  AS      arch/arm/boot/compressed/piggy.o
  LD      arch/arm/boot/compressed/vmlinux
  OBJCOPY arch/arm/boot/zImage
  Kernel: arch/arm/boot/zImage is ready
tair@tair-Bravo-15-C7VE:~/dev/linux$
```

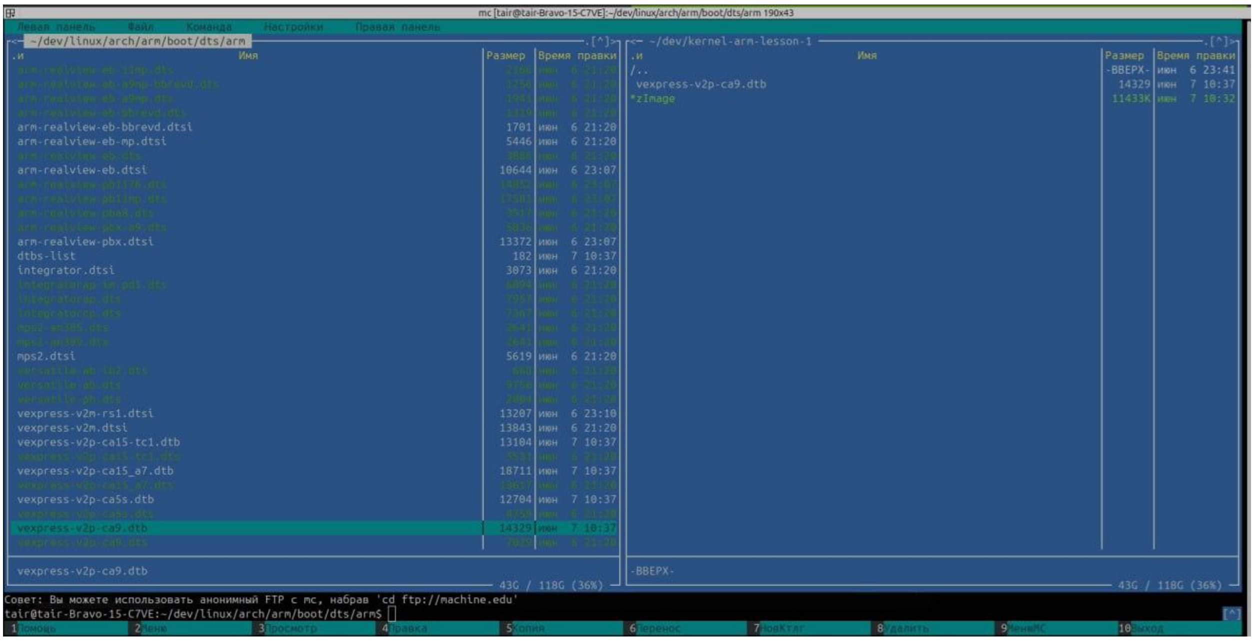


4. Сборка дерева устройств (Device Tree):

Использовалась команда:

ARCH=arm CROSS_COMPILE=arm-linux-gnueabihf- make -j6 zImage

Копирование файла в рабочую директорию:



5. Компилирование файла init и его сжатие:

init.c

```c
#include <stdio.h>
#include <unistd.h>


int main(void)
{
    printf("Hello, Tair!!!!!!!!!!!!!!\n");
    sleep(20);
}
```

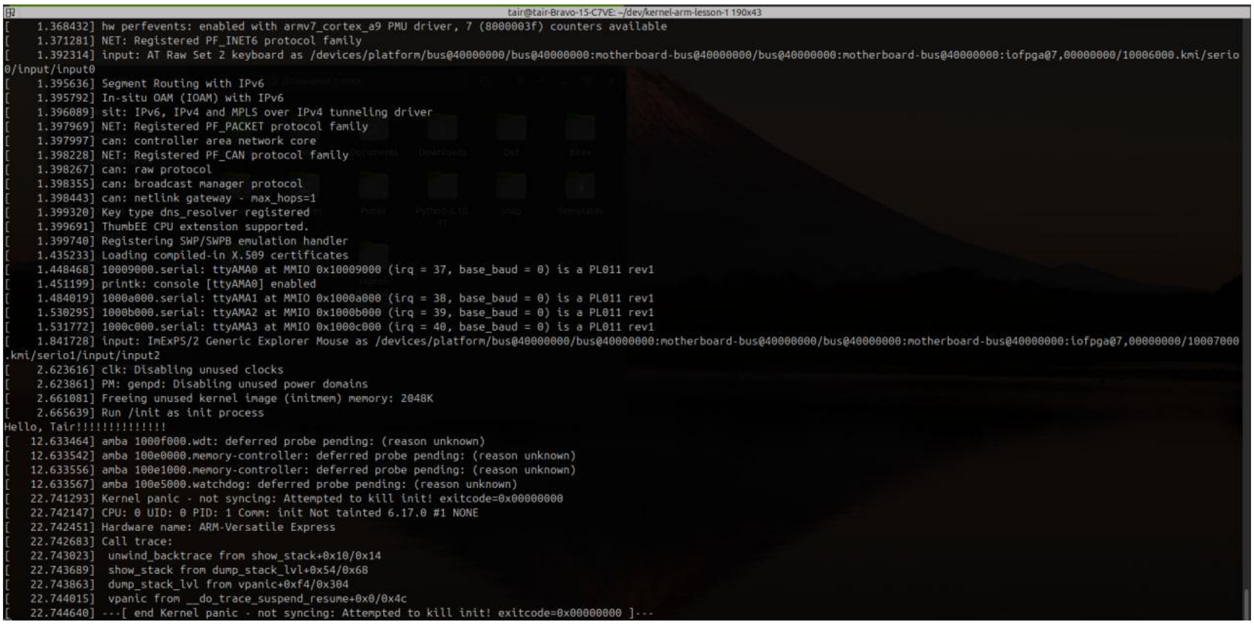
gcc -static init.c -o init

и echo init | cpio -o -H newc | gzip > test.cpio.gz

6. Запуск ядра с файлом init и dtb

QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -initrd test.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0" -nographic

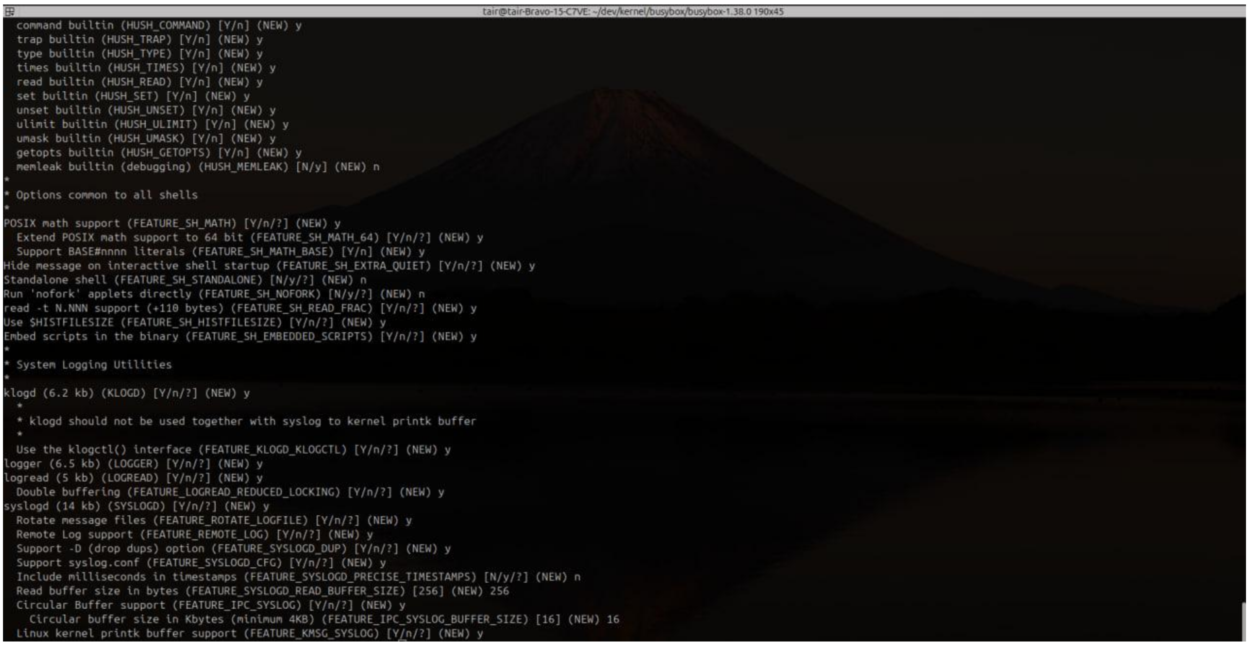
Результат:



Запуск ядра на базе корневой файловой системы busybox

1. Сборка дефолтного конфига:

ARCH=arm make menuconfig

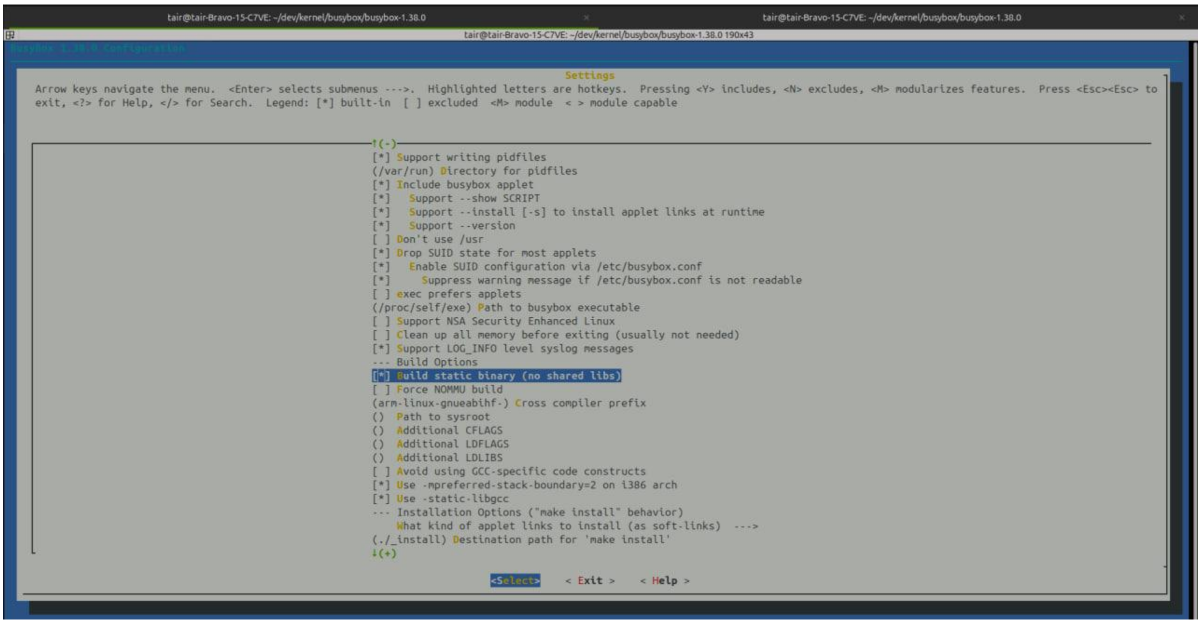


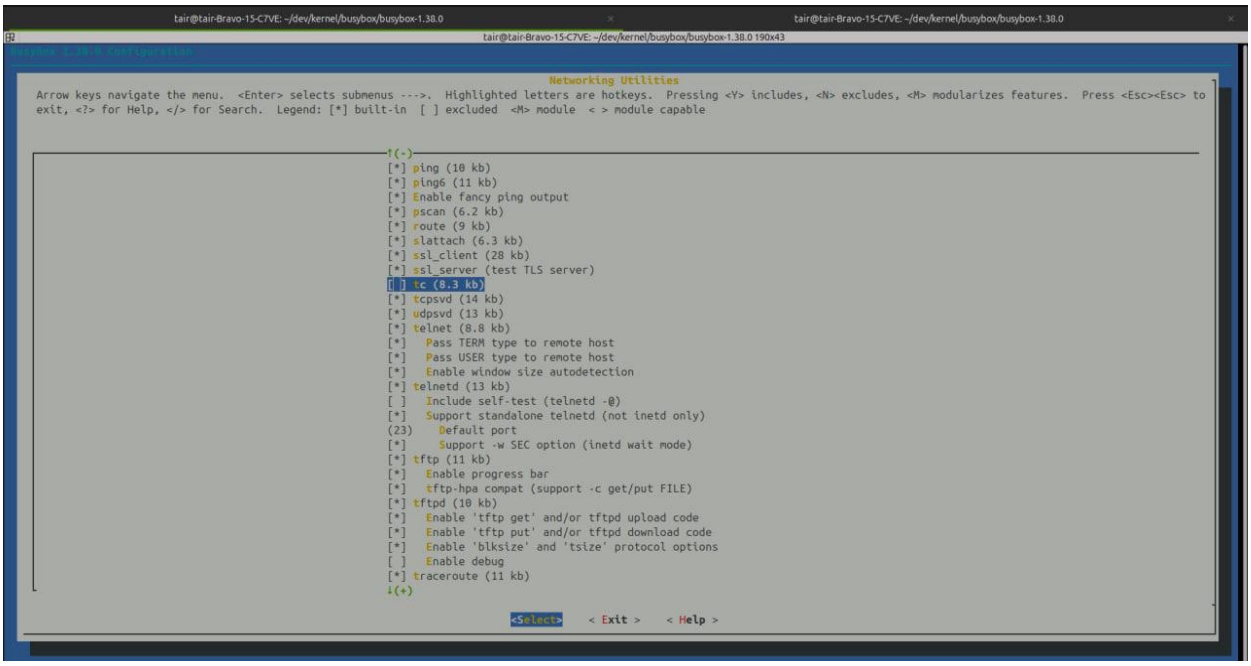
2. Изменение конфигурационного файла:

Были изменены такие параметры:

- Build static binary
- tc (отключил, из-за того, что компилятор выдавал ошибки)
- добавил префикс компиляции

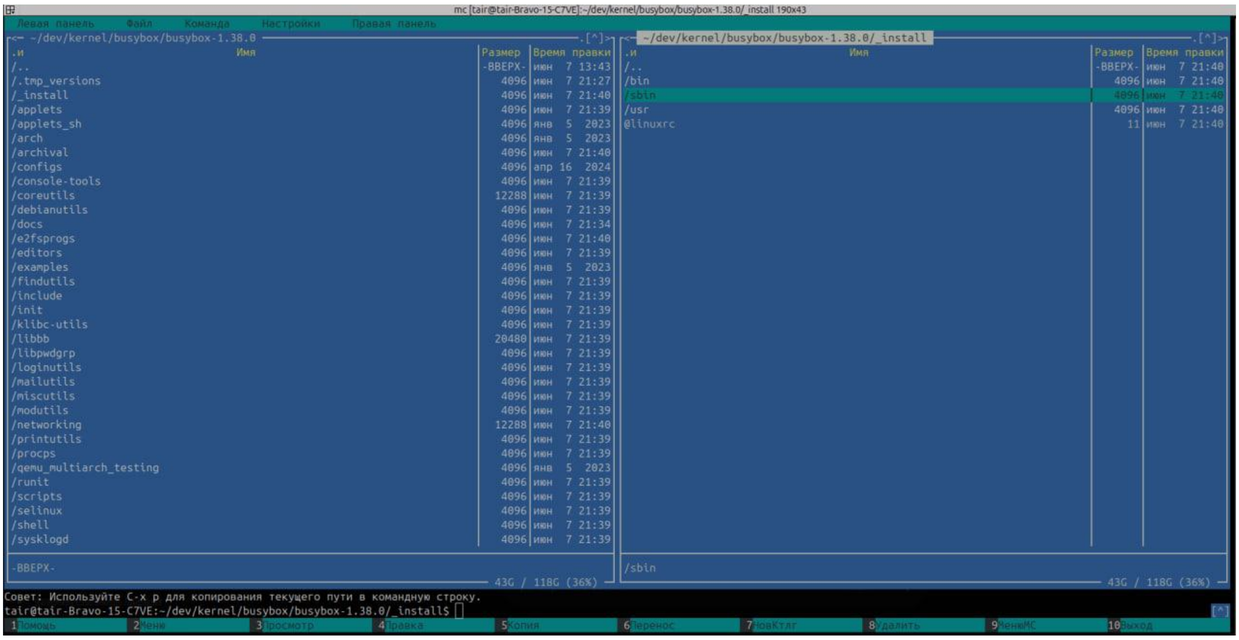
ARCH=arm make menuconfig





3. Сборка файловой системы:

make install



4. Запуск ядра с файловой системой

Перед этим папку _install сжал и поместил в папку рядом с образом

QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0 rdinit=/bin/ash" -nographic